

Math 211 Final Exam

14 May 2018

Questions 1–4: 10 points each, questions 5–8: 15 points each.

1. Use the Divergence Theorem to find the outward flux of the vector field

$$\mathbf{F}(x, y, z) = (x^3 - e^{yz}) \mathbf{i} + (y^3 + \sin z^2) \mathbf{j} + (z^3 - xy) \mathbf{k}$$

over the sphere $x^2 + y^2 + z^2 = z$.

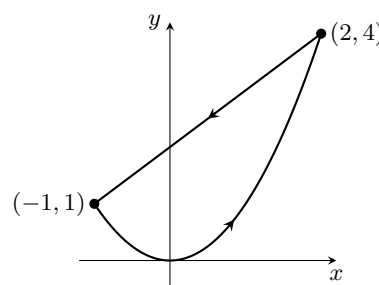
2. Find $\lim_{(x,y) \rightarrow (0,0)} \frac{xy \sin x \cos y}{x^4 + y^2}$ or show that it does not exist.

3. (a) Find a vector function that represents the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the paraboloid $z = x^2 + 2x + y^2$.

- (b) Let C be the curve with vector equation $\mathbf{r}(t) = (\ln t) \mathbf{i} + t^2 \mathbf{j} + 2\sqrt{t} \mathbf{k}$. Find an equation for the normal plane to C at the point $(0, 1, 2)$.

4. Let C be the curve which consists of the arc of the parabola $y = x^2$ from $(-1, 1)$ to $(2, 4)$, followed by the line segment from $(2, 4)$ to $(-1, 1)$. Use Green's Theorem to evaluate

$$\int_C (xy + \arctan x) dx + (x^2 + 3 \ln(1 + y)) dy.$$



5. Use Lagrange multipliers to find the absolute extrema of the function $f(x, y) = x^2 e^y$ on the circle $x^2 + y^2 = 3$.

6. Use the Fundamental Theorem for Line Integrals to evaluate

$$\int_C (yz + 1) dx + (xz + 1) dy + (xy + 1) dz$$

where C is a smooth curve from $(-1, 2, 1)$ to $(2, 0, 1)$.

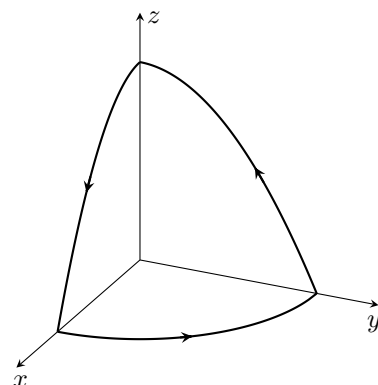
7. Evaluate $\iint_S y \, dS$ where S is the surface with parametric equations

$$x = u^2 - v, \quad y = u^2 + v, \quad z = 2v, \quad 1 \leq u \leq 3, \quad -1 \leq v \leq 1.$$

8. Use Stokes' Theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where

$$\mathbf{F}(x, y, z) = 2x^3 y \mathbf{i} + x^4 \mathbf{j} + (yz + \sqrt{1 + z^3}) \mathbf{k},$$

and C is the boundary of part of the paraboloid $z = 4 - x^2 - y^2$ that lies in the first octant, oriented as shown.



SOLUTIONS

1. Let S denote the given sphere, and E denote the solid region inside S . In spherical coordinates, the given sphere has equation $\rho = \cos \phi$ with $0 \leq \phi \leq \pi/2$. Thus

$$E = \{(\rho, \phi, \theta) \mid 0 \leq \rho \leq \cos \phi, \quad 0 \leq \phi \leq \pi/2, \quad 0 \leq \theta \leq 2\pi\}.$$

Now by the Divergence Theorem we have

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \operatorname{div} \mathbf{F} \, dV = \iiint_E 3(x^2 + y^2 + z^2) \, dV \\ &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^{\cos \phi} 3\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= \frac{\pi}{5} \int_0^{\pi/2} 6 \cos^5 \phi \sin \phi \, d\phi = \frac{\pi}{5} \left[-\cos^6 \phi \right]_0^{\pi/2} = \frac{\pi}{5}. \end{aligned}$$

2. Along the x -axis the limit equals

$$L_1 = \lim_{x \rightarrow 0} \frac{0}{x^4} = 0,$$

and along the path $y = x^2$ it equals

$$L_2 = \lim_{x \rightarrow 0} \frac{x^3 \sin x \cos x^2}{2x^4} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \cos x^2 = \frac{1}{2}.$$

Since $L_1 \neq L_2$, by the two-path rule, the limit does not exist.

3. (a) $\mathbf{r}(t) = \langle \cos t, \sin t, 1 + 2 \cos t \rangle$, $0 \leq t \leq 2\pi$.

- (b) At the point $(0, 1, 2)$ on C , $\ln t = 0 \Leftrightarrow t = 1$. On the other hand, $\mathbf{r}'(t) = \langle 1/t, 2t, 1/\sqrt{t} \rangle$ is tangent to C , hence the normal plane at $(0, 1, 2)$ is normal to the vector $\mathbf{r}'(1) = \langle 1, 2, 1 \rangle$. Therefore, an equation for the normal plane is $1(x-0) + 2(y-1) + 1(z-2) = 0$ which simplifies to $x + 2y + z = 4$.

4. Note that the line joining the points $(-1, 1)$ and $(2, 4)$ has the equation $y = x + 2$. Let

$$P(x, y) = xy + \arctan x, \quad Q(x, y) = x^2 + 3 \ln(1 + y),$$

and let D be the region bounded by the curve C . Then C is positively oriented and by Green's Theorem (verify that all its conditions are indeed satisfied) we obtain

$$\int_C P \, dx + Q \, dy = \iint_D (Q_x - P_y) \, dA = \iint_D (2x - x) \, dA = \int_{-1}^2 \int_{x^2}^{x+2} x \, dy \, dx = \cdots = \frac{9}{4}.$$

5. Let $g(x, y) = x^2 + y^2$. By the method of Lagrange multipliers, the absolute extrema of f on the given circle occur at solutions of the system $\nabla f(x, y) = \lambda \nabla g(x, y)$, $g(x, y) = 3$, that is

$$\begin{cases} 2xe^y = \lambda(2x) \\ x^2 e^y = \lambda(2y) \\ x^2 + y^2 = 3. \end{cases}$$

Multiplying first equation by y and the second equation by x , then subtracting the two resulting equations we obtain $xe^y(2y - x^2) = 0$, which gives $x = 0$ or $x^2 = 2y$.

- If $x = 0$, the last equation gives $y = \pm\sqrt{3}$.
- If $2y = x^2$, the last equation gives $2y + y^2 = 3$ which holds if and only if $y = -3$ or $y = 1$.
Now $y = -3$ is rejected since it yields $x^2 = -6$, while $y = 1$ gives $x = \pm\sqrt{2}$.

Since $f(0, \pm\sqrt{3}) = 0$ and $f(\pm\sqrt{2}, 1) = 2e$, we conclude that the minimum and maximum values of f on the circle $x^2 + y^2 = 3$ are 0 and $2e$ respectively.

6. Let f be a potential function of $\mathbf{F}$. Then $f_x = yz + 1$ which implies $f(x, y, z) = xyz + x + g(y, z)$ for some function g . Now since $f_y = xz + 1$, we must have $xz + 0 + g_y = xz + 1$, which implies $g(y, z) = y + h(z)$ for some function h . Finally, since $f_z = xy + 1$, we must have $xy + h'(z) = xy + 1$ which holds for $h(z) = z$. We conclude that $f(x, y, z) = xyz + x + y + z$ is a potential function for $\mathbf{F}$.

Now by the Fundamental Theorem for Line Integrals we obtain

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(2, 0, 1) - f(-1, 2, 1) = 3.$$

7. Let $\mathbf{r}(u, v) = \langle u^2 - v, u^2 + v, 2v \rangle$. Then

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2u & 2u & 0 \\ -1 & 1 & 2 \end{vmatrix} = 2u \langle 2, -2, 2 \rangle \implies \|\mathbf{r}_u \times \mathbf{r}_v\| = 4\sqrt{3}u \quad (\text{since } u \geq 0 \text{ on } S).$$

Using this we obtain

$$\iint_S y \, dS = \int_1^3 \int_{-1}^1 (u^2 + v) \cdot 4\sqrt{3}u \, dv \, du = 4\sqrt{3} \int_1^3 \int_{-1}^1 (u^3 + uv) \, dv \, du = \dots = 160\sqrt{3}.$$

8. Let S be part of the paraboloid $z = 4 - x^2 - y^2$ that lies in the first octant, oriented upward. Then C is the positively oriented boundary of S . By Stokes' Theorem (verify that all its conditions are satisfied) we have

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}.$$

Note that

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x^3y & x^4 & yz + \sqrt{1+z^3} \end{vmatrix} = \langle z, 0, 2x^3 \rangle,$$

and that S is a graph $z = 4 - x^2 - y^2$ over the region

$$D = \{(x, y) \mid x^2 + y^2 \leq 4, x \geq 0, y \geq 0\} = \{(r, \theta) \mid 0 \leq r \leq 2, 0 \leq \theta \leq \pi/2\}.$$

Using these we obtain

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} &= \iint_D \left[-(4 - x^2 - y^2)(-2x) - 0(-2y) + 2x^3 \right] dA = \iint_D (8x - 2xy^2) dA \\ &= \int_0^{\pi/2} \int_0^2 (8r \cos \theta - 2r^3 \cos \theta \sin^2 \theta) r \, dr \, d\theta = \dots = \frac{64}{3} - \frac{64}{15} = \frac{256}{15}. \end{aligned}$$